

J. Thomas Farrar | Curriculum Vitae

Senior Scientist – Woods Hole Oceanographic Institution

Physical Oceanography Department

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Research Interests

Ocean-atmosphere interaction; dynamics and thermodynamics of the upper ocean; tropical dynamics and equatorial waves; oceanic internal waves and eddies; satellite oceanography; ocean observing and instrumentation.

These interests are pursued using in situ observations, satellite observations, and numerical models to test hypotheses and test or formulate simplified physical models to help us understand ocean dynamics and air-sea interaction.

Education

Ph.D.: Physical Oceanography, Massachusetts Institute of Technology-Woods Hole Oceanographic Institution, 2007

S.M.: Physical Oceanography, Massachusetts Institute of Technology, 2003

B.S.: Physics, University of Oklahoma, 2000

B.A.: Philosophy, University of Oklahoma, 2000

Positions Held

2022-present: Senior Scientist Woods Hole Oceanographic Institution (WHOI), Physical Oceanography Department.

2022-present: Visiting Professor Cornell University, Department of Civil and Environmental Engineering.

2011-present: Adjunct Faculty University of Massachusetts, School of Marine Sciences.

2018-2020: Visiting Scientist Cornell University, Department of Earth and Atmospheric Sciences.

2016-2022: Associate Scientist with Tenure WHOI, Physical Oceanography Department.

2012-2016: Associate Scientist WHOI, Physical Oceanography Department.

2008-2012: Assistant Scientist WHOI, Physical Oceanography Department.

2007-2008: Postdoctoral Investigator WHOI, Physical Oceanography Department.

2004-2007: Graduate Research Assistant WHOI, Physical Oceanography Department.

2003-2004: Research Associate I WHOI, Physical Oceanography Department.

2000-2003: Graduate Research Assistant WHOI, Physical Oceanography Department.

1998-2000: Student Laboratory Assistant Magneto-optical trapping laboratory (atomic-molecular physics), Physics Department, University of Oklahoma.

Selected Academic and Professional Honors

2026: NASA Group Achievement Award for the SWOT Cal/Val Field Campaign Team, for “the oceanographic validation of measurements from the NASA pathfinder mission SWOT, enabling profound new insights into Earth’s climate”.

2024: NASA Group Achievement Award for the Sub-Mesoscale Ocean Dynamics Experiment, for “outstanding achievements in advancing the understanding of small-scale ocean dynamics and their role in the Earth’s climate system”

2021: Science Team Co-Lead and US lead oceanographer for the NASA-CNES-CSA-UKSA Surface Water Ocean Topography (SWOT) satellite mission.

2018: Principal Investigator of the Sub-Mesoscale Ocean Dynamics Experiment (S-MODE), a \$30M NASA Earth Venture Suborbital Mission with 21 co-investigators, remote sensing measurements from three aircraft, and three research cruises that took place in 2021-2023.

2017: American Meteorological Society Nicholas P. Fofonoff Award in recognition of research achievement in physical oceanography, for “insightful analysis of observations yielding a deeper understanding of tropical and upper ocean dynamics, and for generous collaboration and leadership in major field experiments”

2012: Editors’ Citation for Excellence in Refereeing, Journal of Geophysical Research-Oceans

2006: Outstanding Student Paper Award, AGU Ocean Sciences meeting

2000: MIT Presidential Fellowship

2000: Fowler Prize for Most Outstanding Physics Student, University of Oklahoma

1999: Phi Beta Kappa

1999: Golden Key National Honor Society

1998: Sigma Pi Sigma (physics honor society)

1994: National Merit Scholar

National and International Service Activities

- Co-lead, SWOT Science Team (Surface Water Ocean Topography satellite mission), 2021-present
- Principal Investigator of NASA Earth Ventures Suborbital Investigation, S-MODE (Sub-Mesoscale Ocean Dynamics Experiment), 2019-present
- US Lead Oceanographic Subject Matter Expert for SWOT Algorithm Development Team, 2021-present
- Member, Tropical Pacific Observing System Scientific Advisory Committee, 2023-present
- Co-chair, TPOS-2020 (Tropical Pacific Observing System) Task Team on Planetary Boundary Layers, 2015-2020
- Co-chair, SPURS-2 (Salinity Processes in the Upper-ocean Regional Study) Science Steering Committee, 2015-2020
- Co-chair, US CLIVAR Process Study and Model Improvement Panel, 2013-2014
- Organizer, SWOT Science Campaign Workshop, October 2018.
- Organizer, Doppler Scatterometer Field Campaign Workshop, August 2017.
- Organizer, SPURS-2 (Salinity Processes in the Upper-ocean Regional Study) Planning Workshop, April 2014.
- Subject Matter Expert, NASA Surface Water Ocean Topography Algorithm Development Team, 2018-present
- Member of Organizing Committee for the US CLIVAR Workshop on Surface Currents in the Coupled Ocean-Atmosphere System, February 2020
- Member of International CLIVAR Atlantic Implementation Panel, 2011-2015

- Member of US CLIVAR Process Study and Model Improvement Panel, 2012-2016
- Member of TPOS-2020 (Tropical Pacific Observing System) Scientific Steering Committee, 2014-present
- Member of NASA Surface Water Ocean Topography Algorithm Development Team steering committee, 2014-2016
- Member of Science Committee of the European Space Agency's Ocean Salinity Science and Salinity Remote Sensing Workshop, 2014
- Member of Office of Naval Research ASIRI (Air-Sea Interaction Regional Initiative) Steering Committee, 2012-2017
- Member of NASA Ocean Surface Topography Science Team, 2012-2023
- Member of NASA Ocean Vector Wind Science Team, 2010-2018
- Member of Office of Naval Research MISO-BoB (Monsoon Intraseasonal Oscillations in the Bay of Bengal) Steering Committee, 2016-2021
- Member of Office of Naval Research ASTRAL (Arabian Sea Transition Layer) Steering Committee, 2020-present
- Member of SPURS (Salinity Processes in the Upper-ocean Regional Study) Science Steering Committee, 2011-2015

WHOI Committees

- WHOI Physical Oceanography Department Postdoc Mentoring Coordinator, 2024-present
- Member of WHOI Scientific Staff Executive Committee, 2023-2026
- Member of WHOI Ocean Vital Signs Network Vision Committee, 2022
- Member of WHOI Chief People Officer (Human Resources Officer) Search Committee, 2022
- Member of WHOI Physical Oceanography Department Diversity, Equity, and Inclusion Committee, 2021-present
- Member of WHOI Physical Oceanography Faculty Recruitment Committee, 2020-2021
- Member of MIT-WHOI Joint Committee for Physical Oceanography, 2011-2015
- Member of MIT-WHOI Joint Program Graduate Admissions Committee, 2010-2013
- Member of WHOI Search Committee for Physical Oceanography Department Chair, 2010
- Member of WHOI Search Committee for Vice President of Marine Operations, 2008-2009
- Member of WHOI Diversity Committee, 2008-2010
- Member of WHOI Gender Equity Program Advisory Committee, 2004-2006
- Secretary, WHOI Postdoctoral Association, elected representative of the physical oceanography department, 2007-2008

Other Service

- Associate Editor of *Geophysical Research Letters* for special collection, *Science from the Surface Water and Ocean Topography Satellite Mission*, 2024-present
- Associate Editor of *Journal of Geophysical Research: Oceans* for special collection, *Science from the Surface Water and Ocean Topography Satellite Mission*, 2024-present
- Reviewer for *Science*, *Geophysical Research Letters*, *Journal of Physical Oceanography*, *Journal of Geophysical Research*, *Ocean Science*, *Journal of Atmospheric and Oceanic Technology*, *Climate Dynamics*, *Journal of Climate*, *Dynamics of Atmospheres and Oceans*, *Monthly Weather Review*, *Geophysical Research Letters*, *Deep Sea Research*, *Journal of Earth System Science*, *Weather and Forecasting*, *Oceanologia*, *Atmospheric Science Letters*, *Ocean Dynamics*, *Journal of Operational Oceanography*, *Remote Sensing of Environment*, *Oceanography*, *Scientific Reports*, *Journal of Atmospheric Sciences*, *ISPRS Journal of Photogrammetry and*

Remote Sensing, Marine Technology Society Journal, Proceedings of the National Academy of Sciences, PLOS One, IEEE Geosci. Remote Sensing Lett., Journal of Climate, IEEE Trans. Geosci. Remote Sensing, and Bulletin of the American Meteorological Society.

- Proposal reviewer for US National Science Foundation, UK Natural Environment Research Council, NASA
- Panel reviewer for NSF (2010), NOAA (2010), NASA (2021, 2022, 2024, 2025)
- President, Sigma Pi Sigma (physics honor society), Univ. Oklahoma Chapter, 1999-2000

Educational Activities

Ph.D. advisor for: Alec Bogdanoff (MIT-WHOI, Ph.D. awarded 2017; co-advised with Carol Anne Clayson). Dr. Bogdanoff went on to co-found a climate-change adaptation consulting firm in Miami ([Brizaga](#)).

M.S. advisor for: Ersen'S Joseph (Univ. Mass. Dartmouth Mechanical Engineering program; co-advised with Amit Tandon; M.S. awarded 2025).

Thesis Committee Member or Examiner for:

- Ping Zhai (MIT-WHOI, S.M. awarded 2011)
- Ping Zhai (MIT-WHOI, Ph.D. awarded 2014)
- Sudip Majumder (U. Mass., Ph.D. awarded 2014)
- Neeti Neeti (Clark Univ., Ph.D. awarded 2012)
- Sophia Merrifield (MIT-WHOI, Ph.D. awarded 2016)
- Julius Busecke (Columbia University; Ph.D. Awarded 2017)
- Gualtiero Spiro Jaeger (MIT-WHOI, Ph.D. awarded 2019)
- Samuel Levang (MIT-WHOI, Ph.D. awarded 2019)
- Julie Jakoboski (MIT-WHOI, Ph.D. awarded 2019)
- Audrey Delpech (Université de Toulouse, Ph.D awarded 2021)
- Shikhar Rae (Univ. Rochester, Ph.D. awarded 2023)
- Siddhant Kerhalkar (U. Mass., Ph.D awarded 2025)
- Weiguang Roger Wu (MIT-WHOI, Ph.D. awarded 2026)
- Paban Bhuyan (U. Conn., Ph.D. awarded 2026)
- Steven Akin (University of Miami, 2025-present)
- Roland Ovbiebo (Univ. California San Diego, 2025-present)
- Thomas Catapano (MIT-WHOI, 2025-present)
- Debarshi Sarkar (U. Mass., 2025-present)
- James Brice (MIT-WHOI, 2026-present)

Postdoctoral advisor for:

- Shannon Davis (2015-2016); former wind energy research program manager at US Department of Energy
- Viviane Menezes (2016-2017); tenure-track scientist at WHOI
- Michael Schlundt (2017-2019); research scientist at GEOMAR Helmholtz Centre for Ocean Research Kiel
- Uriel Zajackovski (2017-2019); Director of Core Analytics at [WTW](#)
- Cesar Rocha (2018-2020); professor at University of São Paulo
- Seth Zippel (2019-2020); professor at Oregon State University
- Leo Middleton (2023-2024); researcher at University of Gothenburg
- Nihar Paul (2024-2026); professor at Azim Premji University, Bengaluru, India
- Iury Simoes-Sousa (2023-2026); professor at University of Delaware

Undergraduate students advised (internships and summer research projects):

- Erica Rosenblum (summer student from Stony Brook University)
- Sourajit Das (summer student from India Institute of Tech., Kharagpur)

- Weiguang Roger Wu (summer student from Univ. California San Diego)
- Steven Akin (guest student project that became his senior capstone project at Angelo State University)
- Roland Ovbiebo (guest student from Nigeria)
- Ersen'S Joseph (internship and Univ. Massachusetts senior capstone project, Department of Mechanical Engineering)
- Viktoriya Balabanova (internship and Univ. Massachusetts senior capstone project, Department of Physics)

Teaching:

- MIT graduate class 12.805, *Data Analysis in Physical Oceanography* (2016, 2017, 2019, 2020, 2022, 2023; co-taught with Jake Gebbie)
- MIT graduate class 12.870, *Air-sea Interaction* (2023, co-taught with Hyodae Seo; 2025, co-taught with Momme Hell)
- Guest lecturer for MIT graduate class 12.808 *Introduction to Physical Oceanography* (2006-2014).
- Guest lecturer for Cornell undergraduate class EAS 3530 *Physical Oceanography* (2019).
- Guest lecturer for Cornell graduate class CEE 6035 *Case Studies in Coastal and Ocean Processes* (2024, two lectures).

Capacity building:

- Instructor for [Coastal Ocean Environment Summer School in Nigeria and Ghana](#) at the University of Calabar in Calabar, Nigeria (July 2026)
- Advisor/host for Abhijith Raj, recipient of a [POGO-SCOR fellowship](#) for early career scientists from developing nations (2026)
- Participation in collaborative fieldwork with oceanographers in Ghana since 2024, working with faculty and students to increase Ghanaian capacity for collecting state of the art oceanographic and atmospheric data
- Participation in collaborative fieldwork with oceanographers in India since 2012, with numerous visits to Indian oceanographic departments to meet with faculty and students
- Instructor at Physical Oceanography Workshop for early-career scientists at Indian Institute of Technology, Madras (January 2017)
- Instructor for international training course on *Emerging Trends in Ocean Observations and Ocean Data Analysis* at the Indian National Centre for Ocean Information Services in Hyderabad, India, attended by 55 trainees from 7 countries (July 2016)

Member: MIT-WHOI Joint Committee for Physical Oceanography (a graduate program oversight committee concerned with curriculum, required courses, general exam structure, and student welfare), 2011-2015.

Member: WHOI Joint Program Admissions Committee, 2010-2013.

Summer advisor: MIT-WHOI Joint Program student Deepak Cherian (2012).

Visiting graduate students advised: Andrew Wells (2007, Cambridge University, co-advised with Claudia Cenedese); visiting graduate student John Prytherch (2010, University of Southampton, co-advised with Bob Weller)

Publications

Contribution statements use CRediT Contributor Roles Taxonomy

* indicates student or postdoc of Farrar; † indicates other student or postdoc.....

Manuscripts in review.....

[1] S. Rai†, S. Wijffels, H. Seo, **J.T. Farrar**, T. Lee, and R.A. Weller. When sampling mismatches take over: Machine-learning approach to find weather regimes that drive large scatterometer and buoy wind differences.

JGR: Oceans (submitted), 2026. Contributions: analysis design, interpretation, and writing.

- [2] L. Middleton*, A. Y. Shcherbina, and J. T. Farrar. Statistics of curvature in the submesoscale surface ocean (submitted). *Journal of Physical Oceanography* (submitted), 2026. Contributions: review/edit.
- [3] R. Yang, J. Wu, N. Scapin, J. T. Farrar, B. Chapron, S. Popinet, and L. Deike. Multilayer simulation of wind-forced breaking wave fields. *Journal of Fluid Mechanics* (submitted), 2026. Contributions: review/edit, interpretation.

Peer-reviewed articles.....

- [4] V. P. Thangaprakash, M. S. Girishkumar, J. Sreelekha, N. Sureshkumar, S. Shivaprasad, M. Ravichandran, D. Sengupta, J. T. Farrar, C. Lee, E. Pattabhi Rama Rao, S. S. V. S. Ramakrishna, and C. V. Naidu. Investigating near-surface thermohaline structure and barrier layer temperature evolution in the northern Bay of Bengal using seaglider and mooring observations in 2015. *Journal of Geophysical Research: Oceans*, 131:e2025JC023682, 2026. Contributions: field data collection, review/edit.
- [5] M.S. Solari[†], S. Roy, C. Moss, G. Valerio, J. Iungo, J.K. Lundquist, B. Adler, L. Bianco, N. Bodini, E. Cinquino, J.T. Farrar, A. Kirincich, R. Krishnamurthy, S. Letizia, T. Myers, P. Muradyan, M. Pekour, J. Sedlar, L. Soldo, J. Wilczak, and S.F. Zippel. Observation of the marine atmospheric boundary layer's response to a solar eclipse. *Boundary Layer Meteorology*, 192(32), 2026. Contributions: review/edit, field data collection.
- [6] T.L. Schlosser, A.J. Lucas, M. Omand, and J.T. Farrar. Monsoons, plumes, and blooms: intraseasonal variability of subsurface primary productivity in the Bay of Bengal. *Ocean Science*, 22:443–458, 2026. Contributions: review/edit, field data collection.
- [7] N. Scapin[†], J. Wu, J.T. Farrar, B. Chapron, S. Popinet, and L. Deike. Growth rate and energy dissipation in wind-forced breaking waves. *Geophysical Research Letters*, 53(1):e2025GL117901, 2026. Contributions: conceptualization, methodology, review/edit, funding acquisition.
- [8] N. Paul*, M. Mathur, J. Sukhatme, J.T. Farrar, and D. Sengupta. Influx of Bay of Bengal waters and stirring trends in the Arabian sea based on satellite altimetry. *Geophysical Research Letters*, 53(8):e2024GL113884, 2026. Contributions: supervision, review/edit, funding acquisition.
- [9] C. Martin-Blanco[†], N. Scapin, J. Wu, S. Popinet, J.T. Farrar, B. Chapron, and L. Deike. Kinematics of gravity–capillary waves above an evolving underwater current. *Journal of Fluid Mechanics*, 1035:A5, 2026. Contributions: conceptualization, methodology, review/edit, funding acquisition.
- [10] Sarah E. Lang, Melissa M. Omand, Mara A. Freilich, Andrey Y. Shcherbina, J. T. Farrar, Leo Middleton, and Ernesto Rodríguez. Estimating submesoscale-driven particulate organic carbon flux from remote sensing at a dynamic upwelling front. *Journal of Geophysical Research: Oceans*, 131(8):e2025JC023752, 2026. e2025JC023752 2025JC023752.
- [11] A. Kirincich, R. Krishnamurthy, D.D. Turner, B. Adler, N. Agarwal, L.K. Berg, L. Bianco, N. Bodini, M. Chabert d'Hieres, J.T. Farrar, H.J.S. Fernando, B. Gaudet, V.P. Ghate, L. Goldberger, A.O. Gonzalez, E. Hall, E. Hines, G. Hodges, G.V. Iungo, R. Jackson, H. Jiang, A. Kinsella, B. Kosovic, R. Kotamarthi, S. Letizia, S. Lipari, J.K. Lundquist, J. Mirocha, C. Moss, P. Muradyan, T. Myers, R.K. Newsom, J. O'Brien, J.B. Olson, M. Pekour, M. Puccioni, D. Rosencrans, S. Roy, C. Sauvage, J. Sedlar, H. Seo, M. Shams Solari, L. Soldo, S. Stierle, X. Sun, E.J. Thompson, E. Traiger, S. Wharton, J. Wilczak, and S. Zippel. Improving process level understanding of boundary layer winds over the Northeast U.S. Shelf: The third Wind

Forecast Improvement Project (WFIP3) (in press). *Bulletin of the American Meteorological Society*, 2026. *Contributions: review/edit.*

[12] S. Kerhalkar[†], A. Tandon, **J.T. Farrar**, J.A. MacKinnon, T.L. Schlosser, L. Johnson, A.J. Lucas, V. Hormann, and L.R. Centurioni. Modulation of diurnal SST and diurnal warm layer variability by salinity-driven stratification in the Bay of Bengal. *Journal of Physical Oceanography*, 56(1):245 – 266, 2026. *Contributions: review/edit, field data collection.*

[13] S. E. Cravatte, M. A. Balmaseda, P. Browne, M. F. Cronin, C. De Mott, **J. T. Farrar**, Y. Fujii, F. Gasparin, W. S. Kessler, T. Lee, M. Mayer, S. McGregor, M. J. McPhaden, A. Santoso, J. Sprintall, and C. Wen. Missing Western Pacific moorings threaten weather and climate forecasting and research. *Frontiers in Climate*, 8, 2026. *Contributions: review/edit, original text.*

[14] M.C. Buijsman, A.F. Waterhouse, E.D. Zaron, B. Yadidya, C. Xu, V. Whitley, J.O. Wenegrat, J. Wang, A.J. Wallcraft, D. Varma, B.C. Tchonang, O. Siyanbola, J.F. Shriver, V.A. Sheremet, U. Send, K. Raja, K. Polzin, H.E. Ngodock, A.J. Moulin, G. Meiners, A.J. Lucas, M. Lankhorst, J.P. Kuehl, S.M. Kelly, Y. Huang, C. Griffin, J.B. Girton, **J.T. Farrar**, Z. Chen, E.P. Chassignet, M.J. Carrier, A. Bracco, B.K. Arbic, M. Andres, and M.A. Abdulfatai. A multi-agency experiment on internal wave energy, mixing, and interactions and their representation in global ocean models and operational forecasts. *Bulletin of the American Meteorological Society*, 107(1):E158 – E182, 2026. *Contributions: review/edit.*

[15] P. Bhuyan[†], C.B. Rocha, L. Romero, and **J.T. Farrar**. Acoustic Doppler current profiler measurements from Saildrones with applications to submesoscale studies. *Journal of Atmospheric and Oceanic Technology*, 43(2):199 – 212, 2026. *Contributions: conceptualization, methodology, review/edit, funding acquisition.*

[16] G.S. Bhat, V. Jampana, E.P. Rama Rao, S.S. Prasad, S.S. Kumar, D. Sengupta, M. Ravichandran, B.P. Kumar, T.S. Kumar, J.B. Edson, **J.T. Farrar**, A.R. Suryachandra, M.P. Maheshwar, A. Jayakumar, and T.J. Anurose. Year-long eddy covariance surface fluxes from a Bay of Bengal mooring compared with operational Monsoon Mission models. *Current Science*, 130:415–422, 2026. *Contributions: conceptualization, methodology, review/edit, field data collection.*

[17] M.S. Ballard, A. Sanchez-Rios, R.T. Taylor, T.F. Duda, **J.T. Farrar**, H. Simmons, J.D. Sagers, J. Mackinnon, A.J. Lucas, L. Johnson, A. Brakstad, and I. Fer. Hydrographic variability and sound-channel dynamics in the Nordic Seas: Implications for predicting acoustic arrival structure (in press). *Journal of Geophysical Research: Oceans*, 131:e2025JC023781, 2026. *Contributions: review/edit, data collection.*

[18] J. Wu[†], S. Popinet, B. Chapron, **J.T. Farrar**, and L. Deike. Turbulence and energy dissipation from wave breaking. *Journal of Physical Oceanography*, 55(9):1521 – 1534, 2025. *Contributions: review/edit, funding acquisition.*

[19] J. Wang, A. J. Lucas, S. Stalin, M. Lankhorst, U. Send, O. Schofield, L. Kachelein, B. Haines, C. Meinig, R. Pinkel, **J.T. Farrar**, and L. Fu. SWOT mission validation of sea surface height measurements at sub-100 km scales. *Geophysical Research Letters*, 52(11):e2025GL114936, 2025. *Contributions: methodology, investigation, review/edit.*

[20] N. T. Vinogradova, T. M. Pavelsky, **J.T. Farrar**, F. Hossain, and L. Fu. A new look at Earth's water, energy, and climate with SWOT. *Nature Water*, 3:27–37, 2025. *Contributions: original text, review/edit.*

[21] C. Ubelmann, **J.T. Farrar**, B. Chapron, L. Gaultier, L. Gomez-Navarro, M.-H. Rio, and G. Dibarboue. A data-driven wind-to-current response function and application to ocean surface current estimates. *Ocean Science*, 21(6):2915–2928, 2025. *Contributions: conceptualization, methodology, original text, review/edit.*

- [22] H.S. Torres, A. Wineteer, E. Rodriguez, P. Klein, A.F. Thompson, D. Perkovic-Martin, J. Molemaker, D. Hypolite, J. Callies, **J.T. Farrar**, E. D'Asaro, and M. Freilich. Submesoscale eddy contribution to ocean vertical heat flux diagnosed from airborne observations. *Geophysical Research Letters*, 52(2):e2024GL112278, 2025. *Contributions: conceptualization, review/edit, funding acquisition.*
- [23] **J.T. Farrar**, E. D'Asaro, E. Rodríguez, A. Shcherbina, L. Lenain, M. Omand, A. Wineteer, P. Bhuyan, F. Bingham, A. B. Villas Boas, E. Czech, J. D'Addezio, M. Freilich, L. Grare, D. Hypolite, G. Jacobs, P. Klein, S. Lang, I. M. Leyba, Z. Li, A. Mahadevan, J. McWilliams, D. Menemenlis, L. Middleton, J. Molemaker, L. O'Neill, D. Perkovic-Martin, N. Pizzo, L. Rainville, C. Rocha, R. M. Samelson, I. Simoes-Sousa, N. Statom, A. Thompson, D. Thompson, H. Torres, I. Uchoa, J. Wenegrat, and E. Westbrook. S-MODE: the Sub-Mesoscale Ocean Dynamics Experiment. *Bulletin of the American Meteorological Society*, (106):E657–E677, 2025.
- [24] I.T. Simoes-Sousa*, C.M.L. Camargo, J. Tavora, A. Piffer-Braga, **J.T. Farrar**, and T. M. Pavelsky. The May 2024 flood disaster in Southern Brazil: Causes, impacts, and SWOT-based volume estimation. *Geophysical Research Letters*, 52(4):e2024GL112442, 2025. *Contributions: review/edit, supervision, funding acquisition.*
- [25] N. Scapin[†], J. Wu, **J.T. Farrar**, B. Chapron, S. Popinet, and L. Deike. Momentum fluxes in wind-forced breaking waves. *Journal of Fluid Mechanics*, 1009:A20, 2025. *Contributions: conceptualization, review/edit, funding acquisition.*
- [26] S. Rai[†], **J.T. Farrar**, and H. Aluie. Atmospheric wind energization of ocean weather. *Nature Communications*, 16:1172, 2025. *Contributions: review/edit.*
- [27] L. Middleton*, W. Wu, T.M.S Johnston, D. Tarry, **J.T. Farrar**, P. Poulain, T. Ozgökman, A. Shcherbina, A. Pascual, C. McNeill, M. Belgacem, M. Berta, K. Abbott, A. Worden, F. Wittmers, A. Kinsella, L. Centurioni, V. Hormann, E. Cutolo, J. Tintoré, S. Ruiz, B. Casas, H. Chelsack, CALYPSO Collaboration, E. D'Asaro, and A. Mahadevan. Observations of a splitting ocean cyclone resulting in subduction of surface waters. *Science Advances*, 11(30):eadu3221, 2025. *Contributions: supervision, review/edit, funding acquisition.*
- [28] L. Kachelein[†], J. Wang, A. J. Lucas, A. Delpech, **J.T. Farrar**, M. Archer, M. Lankhorst, B. C. Tchonang, U. Send, S. Stalin, J. Sevardjian, and O. Schofield. Sub-100 km ocean processes revealed by structure functions of SWOT sea surface height and in situ observing network. *Journal of Geophysical Research*, 130(11):e2025JC022639, 2025. *Contributions: methodology, investigation, review/edit.*
- [29] L. Grare, L. Lenain, and **J.T. Farrar**. Observing ocean-atmosphere fluxes from autonomous surface vehicles. *Geophysical Research Letters*, 52:e2025GL115335, 2025. *Contributions: conceptualization, methodology, original text, review/edit.*
- [30] H.S. Torres, E. Rodriguez, A. Wineteer, P. Klein, A.F. Thompson, J. Callies, E. D'Asaro, D. Perkovic-Martin, **J.T. Farrar**, F. Polverari, and R. Akbar. Airborne observations of fast-evolving oceansubmesoscale turbulence. *Communications Earth & Environment*, 5, 2024. *Contributions: conceptualization, review/edit, funding acquisition.*
- [31] R. Samelson and **J.T. Farrar**. Models of the sea-surface height expression of the internal-wave continuum. *Journal of Physical Oceanography*, 54:2099–2117, 2024. *Contributions: conceptualization, methodology, software, original text, review/edit, analysis, funding acquisition.*
- [32] L.D. Riihimäki, M. F. Cronin, R. Acharya, N. Anderson, J. Augustine, K. A. Balmes, P. Berk, R. Bozzano, A. Bucholz, K. J. Connell, C. J. Cox, A. G. di Sarra, J. Edson, C.W. Fairall, K. **J.T. Farrar**, Grissom,

- M. T. Guerra, V. Hormann, K. J. Joseph, C. Lanconelli, Meloni D. Melin, F. and, M. Ottaviani, S. Pensieri, K. Ramesh, D. Rutan, N. Samarinas, S. R. Smith, S. Swart, A. Tandon, E. J. Thompson, R. Venkatesan, R. K. Verma, V. Vitale, K. S. Katie S. Watkins-Brandt, R. A. Weller, C. J. Zappa, and D. Zhang. Ocean surface radiation measurement best practices. *Frontiers in Marine Science*, 11, 2024. *Contributions: conceptualization, methodology, review/edit.*
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Farrar, J.T. Barotropic Rossby waves seen radiating from tropical instability waves in the Pacific Ocean. NASA Ocean Surface Topography Science Team Meeting, Seattle, WA. 2009.

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Farrar, J.T., L. Fu, R. Morrow, F. d'Ovidio. A pair of notional SWOT science campaigns. SWOT Science Team Meeting, Montreal, Quebec, June 2018 (Invited).

Farrar, J.T., M. Cronin, Planetary Boundary Layer Task Team Status report. TPOS2020 Steering Committee Meeting, Seattle, WA, Oct 2017.

Farrar, J.T., Do internal waves affect our science goals in ways other than aliasing? Workshop for the Winds and Currents Mission, La Jolla, CA, April 2018.

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Farrar, J.T., and 10 coauthors. Contrasting upper-ocean heat and salt balances and dynamics in SPURS 1 and 2. Ocean Sciences Meeting, Portland, OR, Feb 2018.

Farrar, J.T., Introduction to WHOI and Ocean Remote Sensing. Meeting of National Academies of Science, Engineering, and Medicine Committee on Radio Frequencies, Woods Hole, MA, October 2018 (Invited).

Farrar, J.T., SWOT Science Campaigns: Notional campaigns and science goals. SWOT Science Campaign Workshop, Crystal City, VA, October 2018.

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Farrar, J.T. and coauthors, NASA Earth Venture Suborbital Investigation: Sub-Mesoscale Ocean Dynamics Experiment (S-MODE). SWOT Science Team Meeting, Bordeaux, France, June 2019.

Rocha, C., Wu, W., and **Farrar, J.T.**. Directionality of internal waves. SWOT Science Team Meeting, Bordeaux, France, June 2019.

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Farrar, J.T., et al. Surface current observations in the Sub-Mesoscale Ocean Dynamics Experiment. Ocean Sciences Meeting, San Diego, February 2020 (invited).

Dibarboure, G. and **Farrar, J.T.**, SWOT Low-rate data algorithm and products. SWOT Science Team Meeting, online, September 2021 (Invited).

Farrar, J.T. Equatorial waves across the Pacific, Indian and Atlantic: a virtual poster (or whatever). Ocean Sciences Meeting, February 2022.

Farrar, J.T. Questions About The Global Distribution of Eddy Kinetic Energy. Gordon Research Conference on Ocean Mixing, South Hadley, MA, 2022. (Invited)

Farrar, J.T. Observations of equatorial waves in all three tropical oceans. International Symposium on Geophysical Flows, Indian Inst. Of Technology, Madras, August 2022, 2022. (Invited)

Farrar, J.T. The Sub-Mesoscale Ocean Dynamics Experiment (S-MODE). FilaChange (international conference on ocean processes linking filaments and fine scales with climate change), Providence, RI, 2022. (Invited)

Farrar, J.T. S-MODE and SWOT. Ocean Sciences Meeting, New Orleans, LA, February 2024.

Farrar, J.T. Ocean surface current measurements in the Sub-Mesoscale Ocean Dynamics Experiment (S-MODE). IEEE International Geoscience and Remote Sensing Symposium, Athens, Greece, 2024.

Farrar, J.T. and coauthors, Ocean surface current measurements in the Sub-Mesoscale Ocean Dynamics Experiment (S-MODE). IEEE International Geoscience and Remote Sensing Symposium, Athens, Greece, 2024.

Farrar, J.T. and coauthors, Wave Glider and DBASIS buoy measurements during EKAMSAT 2024 IOP1. Office of Naval Research EKAMSAT meeting, Seattle, WA, 2024.

Farrar, J.T. and coauthors, NORSE Drifting Buoy Air-Sea Interaction System. Office of Naval Research NORSE meeting, Seattle, WA, 2024.

Farrar, J.T. and coauthors, Wave Glider divergence, vorticity, vertical velocity, surface fluxes. 2024 S-MODE Science Team Meeting, Mountain View, CA, 2024.

Farrar, J.T. and coauthors, Observations of tsunami generation by icebergs. AGU Annual Meeting, Washington, D.C., 2024.

Farrar, J.T. and coauthors, Wave Glider divergence, vorticity, vertical velocity, surface fluxes. 2025 S-MODE Science Team Meeting, Mountain View, CA, 2025.

Farrar, J.T. and coauthors, S-MODE KDP-F, NASA Headquarters Key Decision Point – Final Review, Washington, D.C., 2025.

Lectures & Seminars

September 2006: Massachusetts Institute of Technology, Oceanography and Climate Sack Lunch Seminar. *Oceanic mesoscale variability and atmospheric convection on 10° N in the eastern Pacific.*

February 2007 (Invited): National Data Buoy Center. *Buoys and wave-measurement requirements of the WHOI Upper Ocean Processes Group.*

April 2007 (Invited): Lamont-Doherty Earth Observatory, Ocean and Climate Physics Seminar. *Oceanic mesoscale variability and atmospheric convection on 10° N in the eastern Pacific.*

June 2007 (Invited): Geophysical Fluid Dynamics Program Staff Lecture, Woods Hole Oceanographic Institution. *Modulation of the cool skin of the ocean by internal waves* (with C.J. Zappa and C. Cenedese).

July 2007 (Invited): Geophysical Fluid Dynamics Program, Mini-Symposium on "Ocean Bottom and Surface Boundary Layers", Woods Hole Oceanographic Institution. *The ocean's diurnal boundary layer: observations and models.*

October 2007: University of Oklahoma, Department of Meteorology, Seminar series in convection and numerical weather prediction. *Oceanic mesoscale variability and atmospheric convection on 10° N in the eastern Pacific.*

October 2007 (Invited): University of Oklahoma, Department of Physics Colloquium Series. *Planetary-scale equatorial waves in the Pacific Ocean and mathematical analogy to the quantum simple harmonic oscillator.*

November 2007: Woods Hole Oceanographic Institution, Physical Oceanography Seminar. *Observations of equatorial waves in the Pacific Ocean: Dispersion characteristics, meridional sea-level structures, and a previously unobserved wave mode.*

November 2007 (Invited): Oregon State University, Physical Oceanography Seminar. *Observations of the dispersion characteristics and meridional sea-level structure of equatorial waves in the Pacific Ocean.*

November 2007: Oregon State University, Joint Physical Oceanography and Atmospheric Sciences Seminar. *Oceanic mesoscale variability and atmospheric convection on 10° N in the eastern Pacific.*

April 2008: Woods Hole Oceanographic Institution, Physical Oceanography Seminar. *Equatorial and tropical instability waves: synthesis and new observations.*

June 2008: Oregon State University, Physical Oceanography Seminar. *Equatorial and tropical instability waves: synthesis and new observations.*

September 2009: Texas A&M University, Department of Atmospheric Sciences Seminar. *Oceanic mesoscale variability and atmospheric convection on 10° N in the eastern Pacific.*

September 2009 (Invited): Texas A&M University, Department of Oceanography Seminar. *Observations of the dispersion relation and meridional structure of equatorial waves and tropical instability waves.*

October 2009 (Invited): WHOI, Applied Ocean Physics and Engineering Coastal Ocean Fluid Dynamics Laboratory Seminar. *Air-sea exchange in the Red Sea and the role of coastal processes.*

December 2009 (Invited): University of Rhode Island, Graduate School of Oceanography. *Observations of the dispersion relation and meridional structure of equatorial waves and tropical instability waves.*

February 2011 (Invited): Rensselaer Polytechnic Institute, Mathematics Colloquium. *Observations of planetary-scale ocean waves in the equatorial waveguide.*

April 2011 (Invited): University of Massachusetts, Dartmouth, School for Marine Science and Technology. *New observations of equatorial waves and tropical instability waves in the Pacific Ocean.*

June 2011: Woods Hole Oceanographic Institution, Physical Oceanography Seminar. *On the spectrum of equatorial inertia-gravity waves and the significance of zero group velocity* (co-delivered with Ted Durland).

April 2012: Oregon State University, Physical Oceanography Seminar. *Sea-level variability in the tropical Pacific: from inertia-gravity waves to Rossby waves.*

April 2012: University of South Florida, Oceanography Seminar. *New observations of equatorial waves and tropical instability waves.*

May 2012: Woods Hole Oceanographic Institution, Physical Oceanography Seminar. *Sea-level variability in the tropical Pacific: from inertia-gravity waves to Rossby waves.*

April 2013 (Invited): Lamont-Doherty Earth Observatory, Ocean and Climate Physics Seminar. *Sea-level variability in the tropical Pacific: from inertia-gravity waves to Rossby waves.*

March 2016: Woods Hole Oceanographic Institution, Physical Oceanography Seminar. *Long-range teleconnections in the ocean eddy field.*

July 2016 (Invited): National Institute of Ocean Technology, Chennai, India. *Mooring measurements of air-sea interaction at WHOI.*

July 2016 (Invited lectures): International Training Course on Emerging trends in Ocean Observations and Ocean Data Analysis (58 students); Hyderabad, India, Indian National Centre for Ocean Information Services. *Fourier analysis, spectral analysis, and filtering.*

May 2017: Woods Hole Oceanographic Institution, Physical Oceanography Seminar. *Surface velocity measurements from air and space (Doppler scatterometry).*

October 2018: Cornell University, Atmospheric Sciences Seminar. *Ocean-atmosphere interaction and ocean dynamics using measurements from water, air, and space.*

February 2019 (Invited): New York University Courant Institute of Mathematical Sciences. *Long-range Rossby wave radiation from an unstable ocean current.*

March 2019: Cornell University, Atmospheric Sciences Seminar. *Long-range Rossby wave radiation from an*

unstable ocean current.

November 2019: Cornell University, Atmospheric Sciences Seminar. *New observations of equatorial waves.*

April 2021: Scripps Institution of Oceanography, SWOT Interest Group Seminar (virtual). *Rossby wave interference patterns and patchy SSH variance.*

September 2021 (Invited): Oregon State University, Physical Oceanography Seminar. *Remote effects of an unstable ocean current.*

April 2022 (Invited): Nigerian National Association of Geoscience Students (virtual). *Career paths and graduate school in physical oceanography.*

January 2023 (Invited): Cornell University, School of Civil and Environmental Engineering, Environmental Fluid Mechanics Seminar. *Kilometer-scale ocean turbulence from planes, ships, robots and satellites: overview of the NASA S-MODE program.*

February 2023 (Invited): University of Rochester, Department of Mechanical Engineering Seminar Series. *Kilometer-scale ocean turbulence from planes, ships, robots and satellites: overview of the NASA S-MODE program.*

March 2023 (Invited): NASA Ames Earth Science Division Seminar. *Kilometer-scale ocean turbulence from planes, ships, robots and satellites: the NASA S-MODE program.*

November 2024: Cornell University, School of Civil and Environmental Engineering, Environmental Fluid Mechanics Seminar. *Iceberg tsunamigenesis observed from SWOT.*

November 2024 (Invited): University of Rhode Island, Graduate School of Oceanography. *S-MODE: the Sub-Mesoscale Ocean Dynamics Experiment.*

November 2025: Cornell University, School of Civil and Environmental Engineering, Environmental Fluid Mechanics Seminar. *Iceberg tsunamigenesis observed from SWOT (updates!).*

Fieldwork Experience

2025: Chief Scientist for ONR ASTRAL/EKAMSAT cruise, RV *Thomas G. Thompson*, Phuket, Thailand to Phuket, Thailand (Bay of Bengal).

2024: Chief Scientist for WFIP-3 LIDAR buoy deployment cruise, MV *Capt. Joseph E. Pearce*, Woods Hole, MA to Woods Hole, MA.

2019: MISO-BoB cruise, RV *Sally Ride*, Chennai, India to Chennai, India (Bay of Bengal). Led WHOI drogued surface buoy effort.

2017: IRENE cruise, BO *SOCIB*, Palma, Mallorca to Palma, Mallorca. Led WHOI drogued surface buoy effort and Underway CTD sampling.

2016: SPURS-2 deployment cruise in-port work; Honolulu. Worked with group members to assemble surface mooring buoy, prepare instruments and equipment, and load ship.

2012: SPURS deployment cruise, RV *Knorr*; Woods Hole to Azores. Led WHOI mooring effort and Underway CTD sampling.

2010: Chief Scientist for King Abdullah University for Science and Technology mooring recovery cruise, SETE3 (tug) and SETE30 (barge); Durrat, Saudi Arabia to Durrat, Saudi Arabia (Red Sea).

2009: Chief Scientist for King Abdullah University for Science and Technology mooring recovery/redeployment cruise, SETE3 (tug) and SETE30 (barge); Durrat, Saudi Arabia to Durrat, Saudi Arabia (Red Sea).

2008: Chief Scientist for King Abdullah University for Science and Technology mooring deployment cruise, RV *Oceanus*; Jeddah, Saudi Arabia to Jeddah, Saudi Arabia.

2008: Northern Tropical Atlantic Station mooring turnaround cruise, RV *Oceanus*; Woods Hole, MA to Barbados.

2007: CLIMODE and wave-measurement test mooring recovery cruise, RV *Oceanus* (Co-PI with Robert Weller). Participation in planning and execution of a dragging operation to retrieve remnants of a mooring that had previously failed in the Gulf Stream.

2007: Chief Scientist for Wave-measurement Test Mooring deployment cruise, RV *Oceanus*; coastal waters south of Martha's Vineyard, MA.

2003: Chief Scientist for 3 of 5 cruise legs of Coupled Boundary Layers and Air-Sea Transfer Experiment, Low Winds, FV *Nobska* (PI, Robert Weller).

2001: Salt Finger Tracer Release Experiment, RV *Oceanus*; Barbados to Barbados. Supported SF₆ tracer release and microstructure sampling.